

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Examination – Summer 2024

Course: B. Tech.

Branch: Civil Engineering & Allied

Semester: IV

Subject Code & Name: BTCVC406 / BTCIC406 / BTCEC406 Engineering Geology

Max Marks: 60

Date: 26/06/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Draw a suitable diagram wherever necessary.

Marks

Q. 1 Solve Any Two of the following.

- A) Write short notes on: - 6
- a. Meandering
 - b. Tectonic mountains
 - c. Strato-volcano
- B) Write a detailed note on various types of tectonic plate boundaries with neat labelled diagrams. 6
- C) Explain working of river system with various types of drainage patterns. 6

Q.2 Solve Any Two of the following.

- A) Write a short note on: - 6
- a. Gneissose texture
 - b. Batholith
 - c. Cataclastic metamorphism
- B) Write the classification of igneous rocks based on depth, silica and texture. 6
- C) Define sedimentary rock and classification based on grain size and origin. 6

Q. 3 Solve Any Two of the following.

- A) Write a short note on:- 6
- a. Chevron fold
 - b. Graben fault
 - c. Porosity
- B) What is the difference between a confined aquifer and unconfined aquifer? 6
- C) Explain various types of joints and civil engineering significance. 6

Q.4 Solve Any Two of the following.

- A) Write a short note on: - **6**
a. Desk study
b. Rock Quality Designation
c. Core box design and arrangements
- B) Explain trial pit sample collection method with its advantages and disadvantages. **6**
- C) What are the various drilling techniques used for the collection of samples? **6**

Q. 5 Solve Any Two of the following.

- A) Write a short note on: - **6**
a. Tunnel Boring Machine
b. Earthen dam
c. Immersed tunnel
- B) Explain the various favourable and unfavourable geological conditions for dam site. **6**
- C) Explain various types of bridges and their components. **6**

***** End *****